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A machine learning-based new MVA workflow to find correlations in complex datasets applied to fracture diagnostics

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Summary

When attempting to optimize well design to improve fracture effectiveness in unconventional wells for given field, it is often difficult to identify the primary drivers of various fracture characteristics due to the many existing variables such as completion parameters and geologic properties along the wellbore. Additionally, many of the completion or geologic variables may have a nonlinear relationship with the performance of the treatment. This makes finding statistically significant correlations between fracture properties and different variables infeasible using traditional methods. However, in this work, we propose and evaluate a new multivariate analysis (MVA) workflow that utilizes data mining and feature selection techniques for the purpose of identifying and ranking correlations in high-dimensional, nonlinear datasets.

Introduction

There are many different factors that can affect the performance of a hydraulic fracturing job in unconventional reservoirs. Between the heterogeneity of geologic properties along the well and the many completion parameters involved, it is challenging to ascertain which variables, or "features" (which is the terminology most commonly used in the data science field), are more impactful and how each affects the hydraulic fracturing performance (which would be called the "target"). Additionally, many geologic properties and completion parameters are known to have nonlinear relationship with the properties of the induced fractures. However, if the most important features can be identified, the task of optimizing well design becomes a simpler problem.

There are many traditional feature selection methods used in the data science field to identify and rank the features that can accurately predict the target based on perceived correlations. However, when there are dependencies within the features themselves or the features are not linearly related to the target, many of these methods will fail to capture the true feature correlations and may generate misleading results. Therefore, in this study, we set out to develop a more robust method that utilizes data mining methods to accurately capture the effect of variables on a given target.

Dataset Preparation

Two datasets were used in this study: a synthetic dataset and a field dataset. The field dataset used in this study is from a horizonal well located in the unconventional play of Denver-Julesburg Basin (DJ Basin). The well was equipped with fiber optic cable that records both distributed acoustic sensing (DAS) and distributed temperature sensing (DTS) along the 2-mile lateral. The completion design was varied between stages with the intent of using DAS and DTS to estimate fracture characteristics for each stage. The target variable used in the analysis is the hydraulic connectivity of the fractures in the near-wellbore region inferred by the decay rate of tube waves excited by the perforation shot and recorded on DAS. The use of this type of DAS data for fracture diagnostics was first discussed by Schumann and Jin (2020), which showed how tube wave decay rate is related to and correlated with fracture connectivity. Figure 1 shows the methodology used to calculate the tube wave decay rate for each stage. The field dataset includes 25 feature variables and one target variable for 76 stages of the unconventional horizontal well. The 25 feature variables are listed in Table 1. Some of the features are less intuitive than others, so we will explain the meaning of some of them here: Feature 1 refers to the length of time between treatment and the perf shot (to account for partial closure of the fractures); Feature 4 refers to the fluid type (e.g. slick water or cross-link) used and the proppant loading schedule (e.g. step-up or step-down); Feature 15 is initial shut-in pressure (ISIP); Feature 16 is a categorial variables of the different general types of completion designs; Feature 17 indicates the natural fracture density in the stage; Feature 19 is the average temperature in each stage from the DTS warmback data; Feature 20 is the

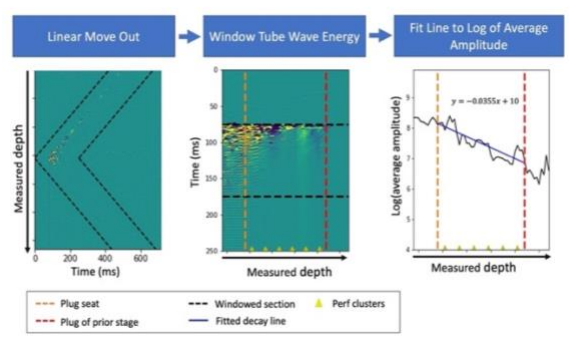

Figure 1 Illustration of the methodology used to calculate the tube wave decay rate for each perf shot in order from left to right.

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number of fracture initiation points per stage (estimated using the DTS warmback); Feature 21 is uniformity index, indicating evenness of flow among the clusters during treatment; Features 23 and 25 represent the heel bias of the proppant and fluid allocation during treatment respectively.

Table 1. List of the 25 features included in the field dataset example.

Feature number	Feature name
1	Hours after treatment
2	Stage spacing (ft)
3	# clusters/stage
4	fluid type
5	bbls/ft
6	Average Pressure (psi)
7	Average Rate (bbl/min)
8	Max Pressure (psi)
9	Max Rate (bbl/min)
10	Proppant Concentration Max (lb/gal)
11	Proppant Concentration Avg (lb/gal)
12	100 Mesh (lbs)
13	30/50 White (lbs)
14	Total Proppant (lbs)
15	ISIP (psi/ft)
16	Design used
17	ALL_NATURAL_FRACTURES
18	GAMMA
19	TEMP31
20	Fips/stage
21	Uniformity Index
22	Proppant total tons
23	Proppant heeward-half tons/total tons
24	Flow total volume
25	Flow heeward-half vol/total vol

In this study, we also used a synthetic dataset with a sample size of 100 to evaluate and compare the performance of different methods, where the ground truth about the relationships between the features and the target was known. The synthetic dataset is generated by the equations written in Python shown in Figure 2. There are eight variables (var1 - var8) and one target. Among the eight variables, var3 is the sum of random noise (between -0.5 and 0.5), var1, and var2; var4 and var5 are both correlated with X_seed, which also consists of random values between -0.5 and 0.5. The target is a non-linear function of five variables (var1 - var5),

```
X_seed = np.random.rand(N,)-0.5

df['var1']=np.random.rand(N,)-0.5
df['var2']=np.random.rand(N,)-0.5
df['var3']=np.random.rand(N,)-0.5 + df['var1'] + df['var2']
df['var4']=np.random.rand(N,)-0.5 + X_seed
df['var5']=np.random.rand(N,)-0.5 + X_seed
df['var6']=np.random.rand(N,)-0.5
df['var7']=np.random.rand(N,)-0.5
df['var8']=np.random.rand(N,)-0.5

df['target'] = np.cos(df['var1'])*np.cos(df['var2']) \
               + df['var1']*df['var2'] + df['var3']/100 \
               + np.sin(df['var4'])*np.sin(df['var5'])
```

Figure 2 Equations used to generate the synthetic dataset example.

whereas the remaining three variables (var6 - var8) do not contribute to the target. With this complexity, correlations between the target and any individual variables would be difficult to identify using traditional visualization tools such as a scatter plot.

Methods

In this study, three methods were compared to understand feature correlations and rank feature importance: 1) Filter method, 2) Embedded method and 3) the MVA method developed here.

Filter method

Filter methods are used to select the subset of relevant features but do not incorporate machine learning algorithms, which makes them run extremely fast (Vikash, 2020). The most commonly used filter methods include Correlation, Chi-square test, ANOVA (Analysis of Variance), Information Gain, etc. In this study, Pearson Correlation was used, which measures linear relationship of two variables or between a variable and the target. It also tells whether a variable is positively or negatively related with the target.

Embedded method

In Embedded methods, feature selection is integrated as part of the learning algorithm, such as Random Forest, Lasso Regression, Ridge Regression, Elastic Net, etc. Embedded methods try all possible ways of splitting features and choose the one that best contribute to the accuracy of the model. As such, Embedded methods are more accurate than Filter methods (Mladenić, 2005). Random Forest Regression method was used to select the top important features in this study. Random Forests build many individual decision trees. Every node in the decision tree is a condition of a single feature, aiming to divide the dataset into two with similar response values end up in the same sub-set. Variance reduction is used to measure the quality of the split. The importance of a feature is computed as the normalized total reduction of variance brought by that feature.

MVA Method

We also propose a new method for feature selection and relative importance ranking. This method, which utilizes a variation of the Sequential Forward Selection algorithm, begins with training a machine learning model (in this case, a Random Forest Regressor) on each feature individually and assessing the performance using 10-fold cross-validation. The top 5 scoring features are then kept, and each is evaluated again in combination with all other features one at a time. Of all these new feature pairs, the top 5 scoring pairs are kept and evaluated in combination with all individual features not in the pair. The top 5 scoring feature trios are then kept, and this process is repeated until either all features are used, or a specified number of features is reached. The best number of features to select can be found

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by examining the trend of the scores at each step. The feature importance can be found by seeing how many times a given features appeared in the top scoring feature combinations at that step. Since a degree of randomness will exist from both the cross-validation and model training, we iterate the entire method from start to finish numerous times to ensure the score trend and feature importance ranking becomes stable.

Results

Filter method

Pearson Correlation matrix is an intuitive tool to visualize and better understand the dataset, as illustrated by Figure 3. The top correlated feature pairs are (var1, var3), (var2, var3) and (var4, var5), which are consistent with the equations listed in Figure 2. The last row/column of Figure 3 shows the correlation coefficient between each feature and target. It reveals the most correlated feature to the target is var4, with a correlation coefficient of 0.19, followed by var6, which has an absolute coefficient value of 0.16. However, var6 has no true relationship with the target, whereas features like var1, var2, var3 are non-linearly correlated with the target. This reveals the limitation that Pearson Correlation has non-linear relationships.

Embedded Method

Figure 4 shows the feature importance ranked by Random Forest. As expected, the top four features are var4, var1, var2 and var5. Different from the original dataset, 1) the importance of var5 is almost half of var4 importance; 2) var3 has a lower importance rank than var6 and var7, both of which do not contribute to the target. This is due to the correlation of these features: var5 is correlated with var4, while var3 is a function of var1 and var2.

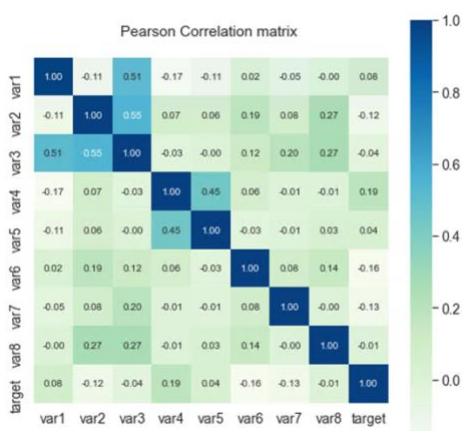

Figure 3 Pearson Correlation matrix for the synthetic data, which reflects the linear relationship between features and the target.

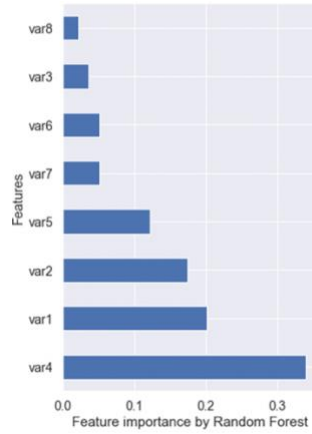

Figure 4 Feature Importance ranked by Random Forest.

The built-in variance reduction mechanism in the Random Forest Regressor can explain the mismatch with the original dataset. When a dataset has two correlated features, such as var4 and var5, either of them can be used as the first splitter, with no preference of one over the other. But once var4 is used, the importance of var5 is significantly reduced because the variance that var5 can diminish has already been lessened by var4. As a consequence, the model thinks var5 has a lower importance. This is not an issue for a prediction problem, because 1) prediction performance won't be affected when deleting one of two features that have similar information 2) it can save computational time. Nevertheless, if we want to do data mining and better understand data correlation, it can lead to an incorrect conclusion that one feature is more important than the other, although in reality they are equally important to the target.

MVA Method

A process of "target shuffling" was applied to the true synthetic dataset and to generate a so-called randomized dataset, where the input features remain same with the true synthetic dataset, but their corresponding targets are shuffled and randomly assigned. This breaks the relationship between input features and targets without changing the dataset. Using the same method, if the true dataset shows a better result than the shuffled data, we can conclude this method well fits the true dataset and is persuasive. (Elder, 2014)

In this study, 100 models for both true and randomized datasets were run with a purpose of getting stable statistical results. Figure 5 shows that R^2 score of the original dataset is always higher than the randomized dataset, demonstrating the statistical accuracy of data mining results. Furthermore, for true dataset (blue box in Figure 5), models with 5 features have the highest R^2 score of 0.6, indicating of best

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performance. Among true synthetic data models with 5 features, if we select those with top 10% performance, then sum up the occurrence of each feature, Figure 6 is obtained. The horizontal axis of the bar chart shows the total times that each feature occurs in these models. Consistent with the original dataset, the top five features are var1 – var5; var4 and var5 have equal importance; var3 has much higher importance rank than features (var8, var7 and var6) that do not contribute to the target. That is to say, better than Random Forest method, MVA method is able to capture the importance of all of the features despite the interactions amongst these features.

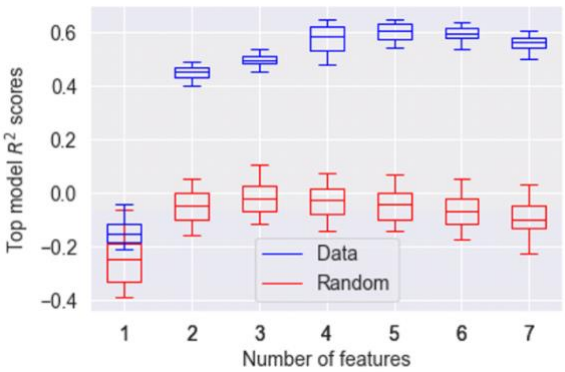

Figure 5 R^2 score of models generated using the new MVA method. The method was run using the true synthetic data (blue) as well as a dataset where the target was randomly shuffled (red) to ensure the results were statistically meaningful.

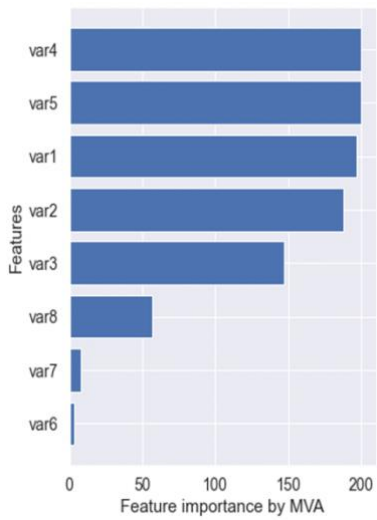

Figure 6 Feature Importance ranked by MVA method.

Application in Field Data

The next step is to apply the MVA method to the field data. The results of 300 models were collected, as displayed in Figure 7. The top six features are total proppant amount, completion fluid type, completion design, max injection rate, total flow volume, and Gamma. These features' importance is significantly higher than the others, indicating they should receive more attention when designing hydraulic fracturing jobs.

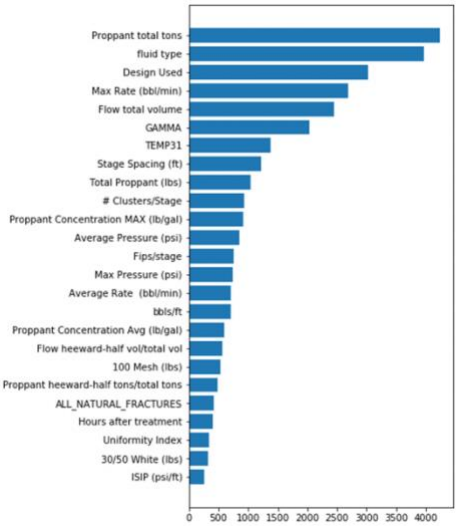

Figure 7 Feature importance ranked by MVA method.

Conclusions

This study uses a synthetic dataset to evaluate and compare the performance of three feature ranking methods. The MVA method developed here is seen to stand out because it is able to correctly capture features' effect on the target. The MVA method was then applied on a field dataset. The results show that the most impactful features to fracture effectiveness are total proppant amount, completion fluid type, completion design, max injection rate, total flow volume and lithology. These are top variables that the operator should focus on when designing stimulation of unconventional wells.

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